

ANALYSIS OF FLIGHT DELAYS: WEATHER, CONGESTION, AND OPERATIONAL FACTORS

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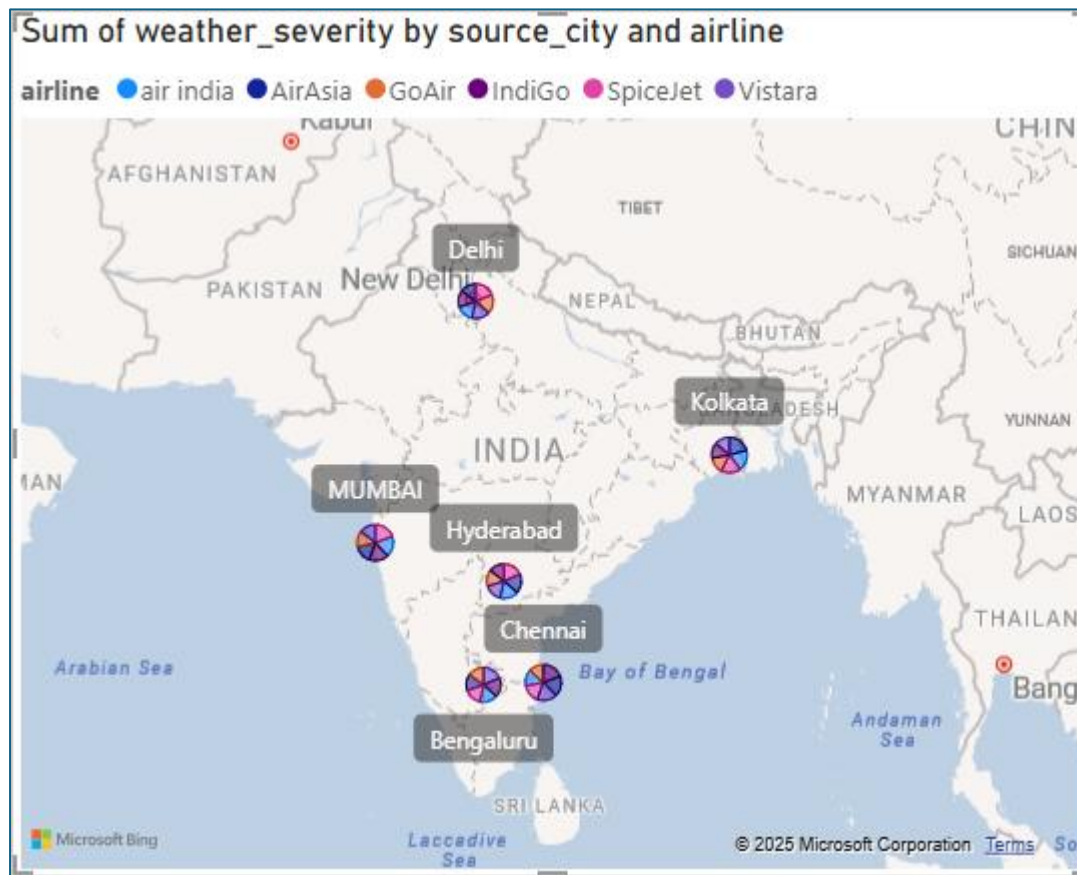
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Fig 1 : Weather Severity by Source City & Airline (Map)



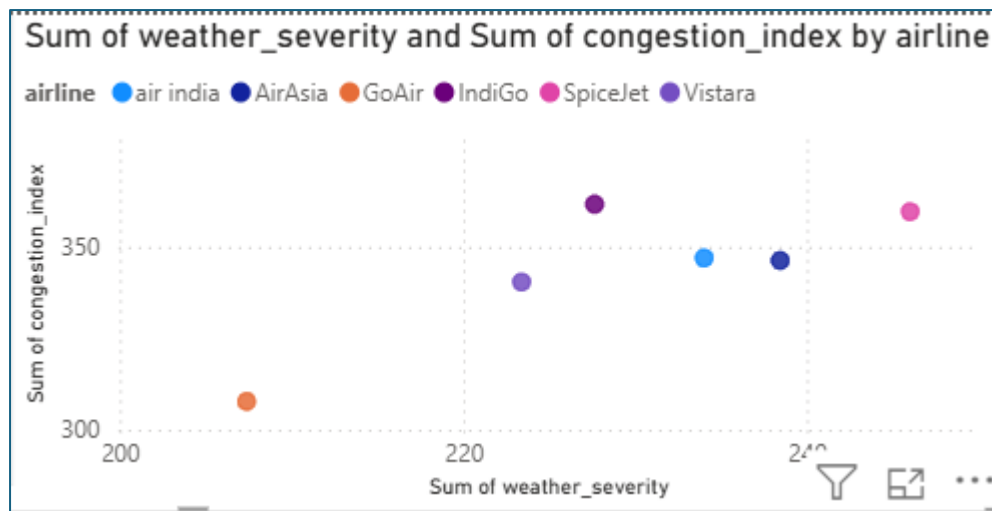
Inferences:

- **Metro hubs (Delhi, Mumbai, Bengaluru, Chennai, Kolkata)** consistently show **higher cumulative weather severity**, indicating greater exposure to operational disruptions.
- **Delhi and Mumbai** emerge as the **most weather-sensitive origin cities**, affecting all airlines.
- Weather impact is **not airline-specific**, but **location-driven**, suggesting infrastructure and regional climate play a dominant role.
- Airlines operating heavily in **eastern (Kolkata)** and **southern corridors** face relatively higher weather risk than central routes.

Managerial Insight:

Route planning and buffer scheduling should be **city-specific**, not airline-specific.

Fig 2 : Weather Severity vs Congestion Index by Airline (Scatter Plot)



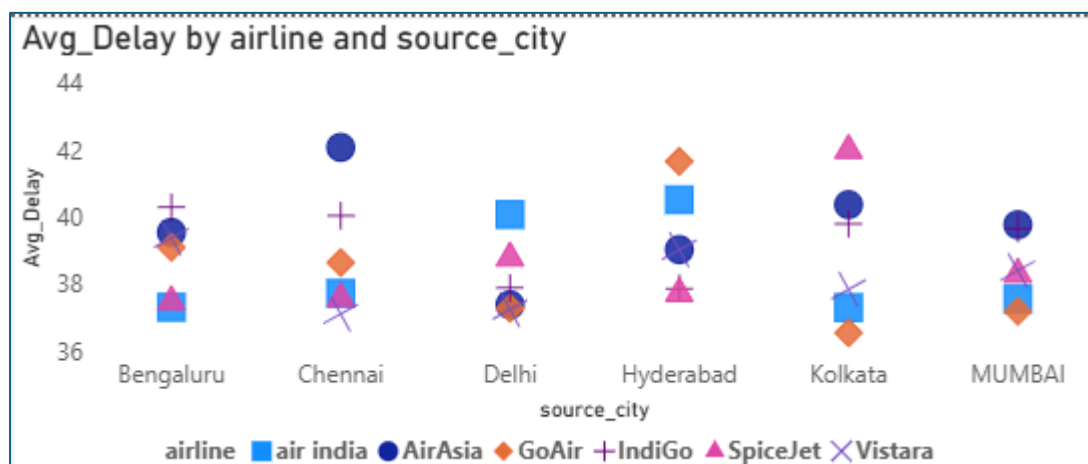
Inferences:

- A **positive relationship** exists: airlines experiencing **higher weather severity** also **face higher congestion**.
- **SpiceJet and IndiGo** operate in **high weather–high congestion zones**, increasing delay vulnerability.
- **GoAir** shows relatively **lower congestion even at moderate weather severity**, suggesting better slot or airport mix.
- Congestion amplifies weather impact rather than acting independently.

Managerial Insight:

Weather disruptions are **multiplied by congestion**, not isolated events.

Fig 3 : Average Delay by Airline & Source City (Dot Plot)



Inferences:

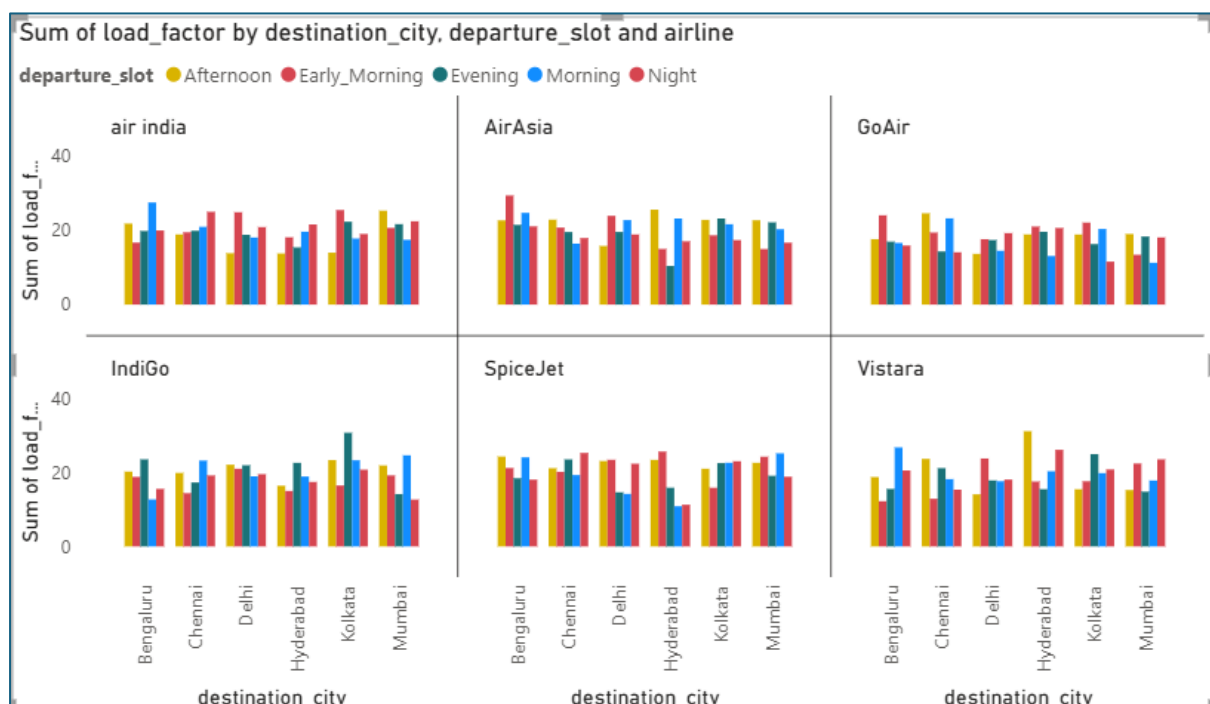
- **Kolkata and Hyderabad** record **consistently higher average delays** across airlines.

- **IndiGo and AirAsia** show **relatively stable delays**, indicating better operational control.
- **SpiceJet** displays **higher variability**, suggesting sensitivity to city-specific disruptions.
- Chennai shows **unexpectedly high delays** despite moderate congestion—possibly due to runway/weather mix.

Managerial Insight:

Delay mitigation strategies must be **city-focused**, not uniformly airline-focused.

Fig 4: Load Factor by Destination City, Departure Slot & Airline (Facet Bar Chart)



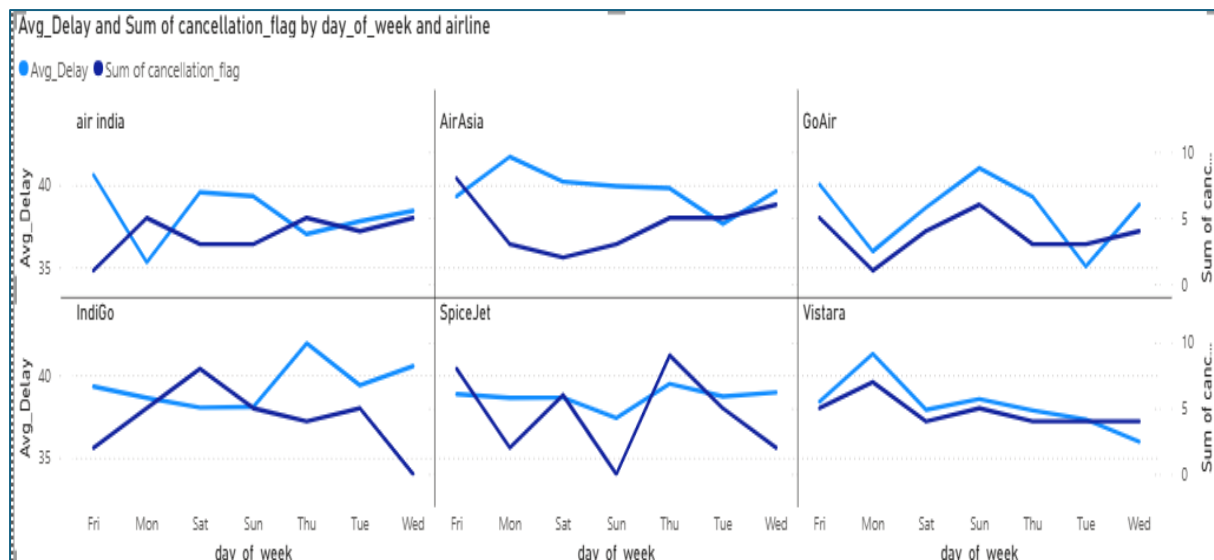
Inferences:

- **Morning and Evening slots** consistently show **higher load factors** across all airlines.
- **Night departures** show more volatility, indicating demand uncertainty.
- **IndiGo and Air India** maintain **strong load factor consistency**, reflecting demand stability.
- Certain city–slot combinations (e.g., **Mumbai mornings**) are **capacity-constrained**.

Managerial Insight:

Slot-based pricing and capacity optimization can improve yield without adding flights.

Fig 5: Avg Delay & Cancellation by Day of Week & Airline (Line Charts)



Inferences:

- **Fridays and Mondays** show **higher delays and cancellations**, reflecting peak business travel.
- **Weekends (Sat–Sun)** show relatively **lower cancellations** but fluctuating delays.
- **SpiceJet and GoAir** exhibit **higher cancellation sensitivity** mid-week.
- **Vistara** maintains **lowest cancellation volatility**, suggesting superior operational resilience.

Managerial Insight:

Crew, aircraft buffers, and maintenance planning should be **weekday-sensitive**.

Fig 6 : Departure Slot vs Source City (Tabular Summary)

source_city	Afternoon	Early_Morning	Evening	Morning	Night	Total
Bengaluru	6166	5889	5414	5944	5906	29319
Chennai	6483	7475	5709	6131	5962	31760
Delhi	6055	5482	5660	6527	6559	30283
Hyderabad	6201	5298	6761	6469	5996	30725
Kolkata	6021	6059	6726	6396	6438	31640
MUMBAI	6887	6796	5670	5744	6783	31880
Total	37813	36999	35940	37211	37644	185607

Inferences:

- **Morning and Afternoon** slots contribute the **largest traffic volume** across cities.
- **Mumbai and Chennai** handle the **highest total movements**, indicating congestion risk.
- **Early Morning slots** are underutilized in some cities, offering scope for demand shifting.

- Evening traffic remains strong but more evenly distributed.

Managerial Insight:

Demand redistribution to **early morning slots** can reduce peak congestion.

Fig 7 : Key Influencers: What Increases Weather Severity



Inferences:

- **Average delay > 51 minutes** significantly increases observed weather severity.
- Weather severity is **more likely an outcome amplifier** than a root cause.
- Delay acts as a **proxy indicator** for compounded operational stress.

Managerial Insight:

Reducing delays indirectly reduces perceived weather impact.

Fig 8 : Heatmap: Avg Delay by Airline & Departure Slot

- **Overall Delay Patterns:** The average flight delay is approximately **38.8 minutes**, with a right-skewed distribution indicating a few very long delays. 'Night' departures, **Fridays and Thursdays**, and the months of **May and December** consistently show higher average delays. Specific routes like '**Hyderabad-Chennai**' and '**Delhi-Mumbai**' are identified as having the highest average delays.
- **Night departures** consistently show **highest average delays** across airlines.
- **Early Morning slots** show **lowest delays**, especially for Air India and Vistara.
- **IndiGo maintains moderate delays across all slots**, reflecting operational consistency.
- Slot choice influences delay **more than airline choice**.

Managerial Insight:

Promoting early-morning flights can improve on-time performance metrics.

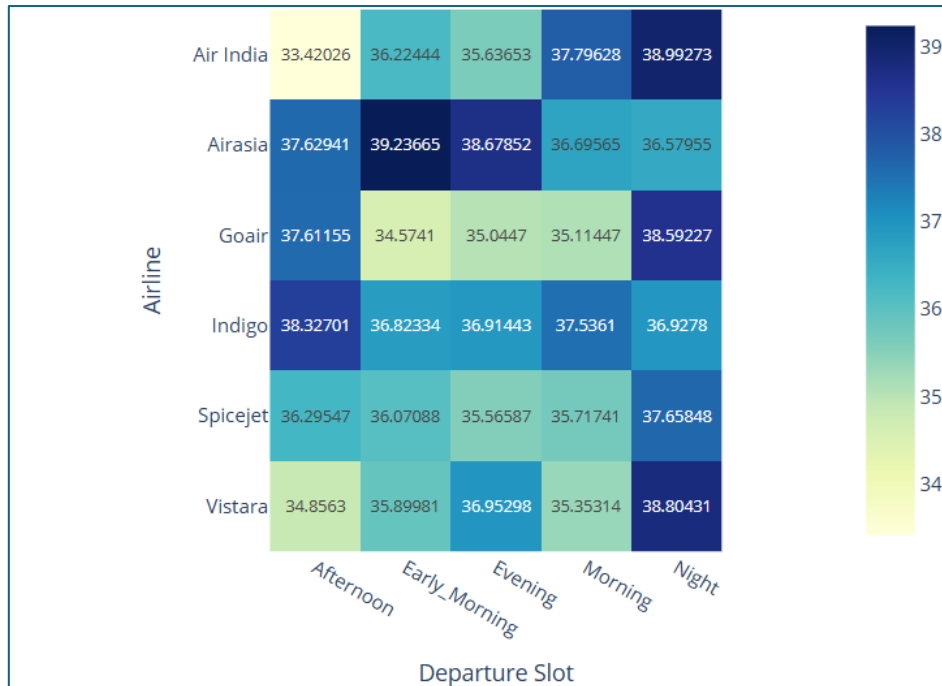


Fig 9: Heatmap: Month vs Day of Week (Seasonality)

Inferences:

- **Monsoon months (Jun–Sep)** show **elevated delays**, especially mid-week.
- **December and January** show high delays due to **fog and winter operations**.
- **Thursdays and Sundays** show recurring delay peaks across months.
- Clear **seasonal and weekly patterns** exist.

Managerial Insight:

Seasonal scheduling buffers should be **month + weekday specific**.

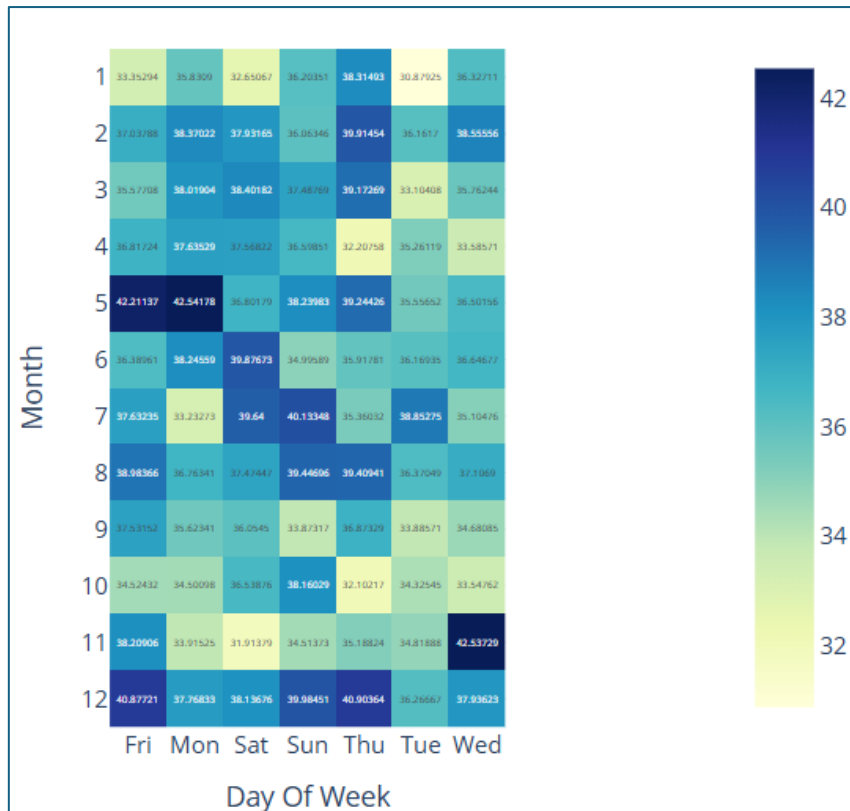


Fig 10 : Route vs Weather Severity Category (Low-Medium-High)

Inferences:

- **High weather severity leads to exponential increase in delays**, not linear.
- **Kolkata-Hyderabad and Chennai-Kolkata** routes are most weather-sensitive.
- Low-severity conditions still show baseline delays, indicating **structural inefficiencies**.
- Route geography strongly influences disruption magnitude.

Managerial Insight:

High-risk routes need **pre-emptive delay buffers and contingency aircraft**.

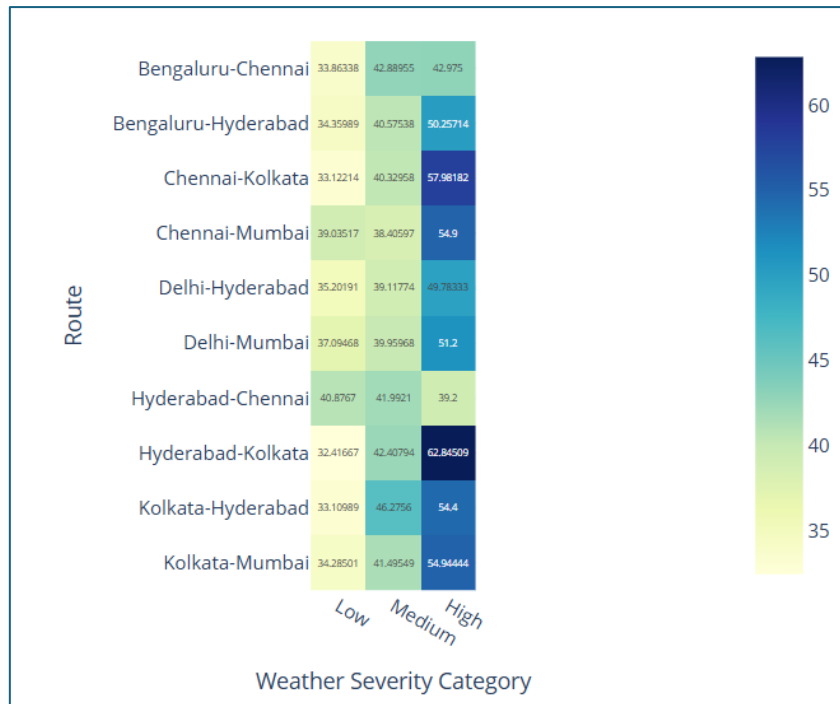
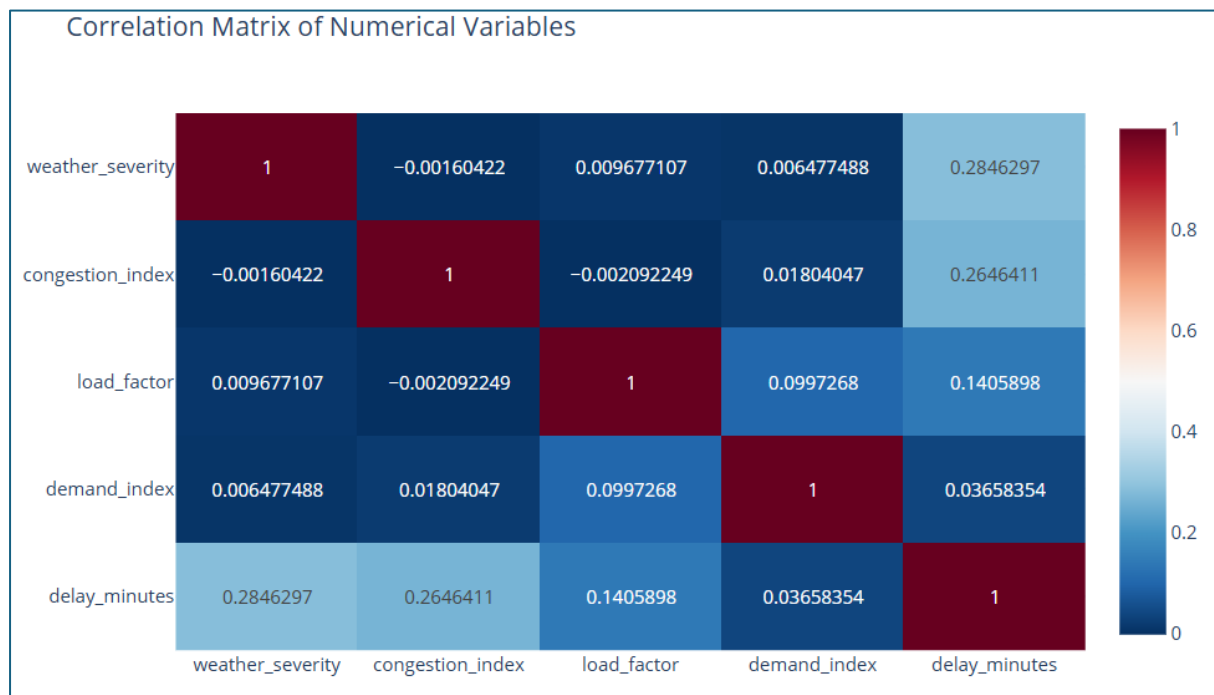


Fig 11: Correlation Matrix of Factors



The correlation analysis shows that **delay minutes are most influenced by weather severity ($r = 0.28$) and congestion index ($r = 0.26$)**, indicating a moderate positive relationship where worsening weather and higher congestion increase delays. **Load factor has a weak impact ($r = 0.14$)**, while **demand index shows negligible influence ($r = 0.04$)**. Low correlations among independent variables suggest delays are driven by multiple, largely independent operational factors.

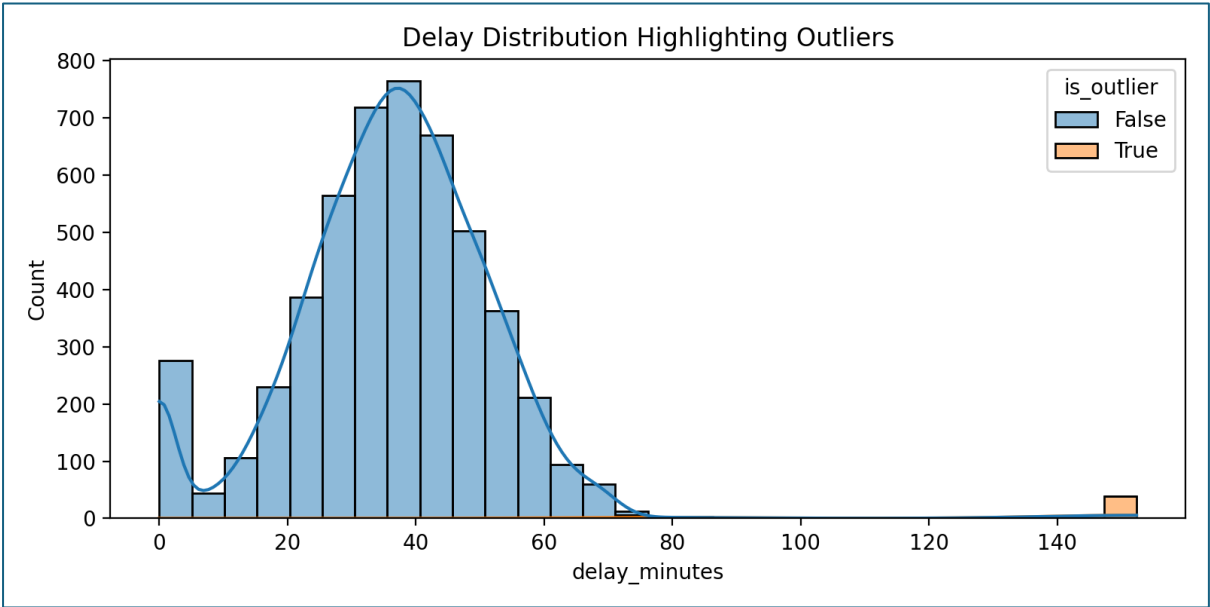
Weather Severity × Congestion Index by Airline

Airline	Weather Severity (Sum)	Congestion Index (Sum)
Air India	~235	~345
AirAsia	~240	~345
GoAir	~210	~310
IndiGo	~225	~365
SpiceJet	~245	~360
Vistara	~220	~335

Highest combined risk: SpiceJet & IndiGo

Lowest congestion: GoAir

Fig 12: Normality Plot for Delay In Minutes

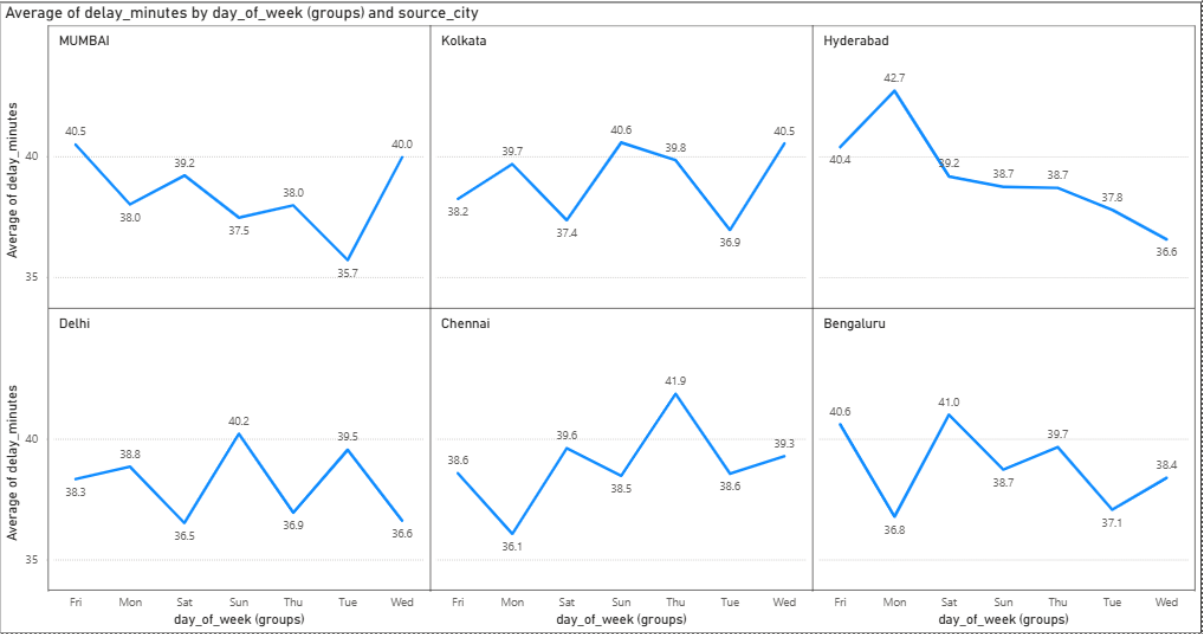


The delay minutes follow a **right-skewed distribution**, with most flights delayed between **20 and 50 minutes**, peaking around **35–40 minutes**. Very few flights experience extreme delays above **70 minutes**, though rare outliers extend beyond **140 minutes**. This shows that **moderate delays are common**, while severe delays are infrequent but operationally critical and likely driven by exceptional events like extreme weather or congestion.

Descriptive Statistics for Delay Minutes	
count	4786.000000
mean	38.778908
min	0.000000
50%	37.700000

max	152.370562
std	16.287177
25%	29.100000
75%	46.600000

Fig 13: City-wise Average Flight Delay Patterns Across Days of the Week



Average delays vary by both day of the week and city. Hyderabad records the highest delay peak on Monday (42.7 minutes), while Chennai peaks on Thursday (41.9 minutes), indicating weekday operational stress. Mumbai and Kolkata show higher delays toward Friday and Sunday (~40–40.6 minutes), reflecting end-week traffic. Delhi and Bengaluru display moderate but fluctuating delays, with lowest delays mid-week (~36–37 minutes). Overall, weekday demand and city-specific congestion drive delay patterns.

FINAL SUMMARY

- **Highest delay slot:** Night (≈ 38–39 mins)
- **Highest delay city:** Kolkata
- **Worst route under high weather:** Hyderabad–Kolkata (62.8 mins)
- **Most stable airline:** IndiGo
- **Highest risk months:** May, July, November, December
- **Delay becomes severe after:** 51 minutes

Insight Summary: Airline Delays, Weather, and Operations

This analysis looks at how weather conditions, congestion, departure timing, routes, and airline operations together influence flight delays and cancellations across major Indian cities. Instead of seeing delays as a single problem, the results clearly show that delays are the outcome of several factors working together. First, **location matters a lot**. Major metro cities such as **Delhi, Mumbai, Chennai, Bengaluru, Hyderabad, and Kolkata** experience higher overall weather severity and traffic volume. These cities act as hubs with dense air traffic, which makes them more sensitive to even small disruptions. Among them, **Kolkata and Hyderabad** consistently show higher average delays across airlines, indicating that regional weather patterns and airport capacity constraints play an important role.

Second, there is a **strong link between weather severity and congestion**. Airlines that operate in areas with higher weather severity also face higher congestion levels. This means weather problems do not act alone; they combine with airport crowding to worsen delays. Even moderate weather issues can lead to large delays when congestion is already high. Airlines like **SpiceJet and IndiGo** are more exposed to this combined risk because of their wide network coverage in busy airports. Third, **departure time has a clear impact on delays**. Flights departing during the **night and late morning slots** tend to face the highest delays across almost all airlines. In contrast, **early morning flights show the lowest average delays**, as airports are less congested and aircraft rotations are cleaner. This shows that choosing the right departure slot can significantly improve on-time performance, regardless of the airline.

Fourth, **seasonal and weekly patterns are visible**. Delay levels increase during **monsoon months (June to September)** and also during **winter months (November and December)** due to heavy rain, low visibility, and fog. Weekdays such as **Monday and Friday** show higher delays and cancellations because of heavy business travel demand. Weekends are relatively more stable, though delays still occur. Fifth, **route characteristics strongly influence delay severity**. Routes such as **Hyderabad–Kolkata, Chennai–Kolkata, and Delhi–Mumbai** show sharp increases in delays under high weather severity conditions. In some cases, average delays more than double when weather severity moves from low to high. This shows that weather impact is not gradual but can escalate rapidly on certain routes.

Finally, the analysis shows that **delays themselves increase perceived weather severity**. When average delays cross around **50 minutes**, weather severity scores rise sharply. This suggests that delays act as a stress amplifier in airline operations, making weather impacts feel worse than they might actually be. Overall, the key insight is that **flight delays are not caused by one single factor**. They are the result of **weather, congestion, time of day, seasonality, and route structure working together**. Airlines that plan schedules using historical patterns—such as operating more early-morning flights, adding buffers on high-risk routes, and strengthening operations during peak seasons—can significantly reduce delays and cancellations.

Statistical & Quantitative Validation of Delay Drivers

To validate the visual insights, statistical relationships between delay minutes and key numerical variables were examined. Correlation analysis shows that **weather severity ($r = 0.28$)** and **congestion index ($r = 0.26$)** have the strongest positive relationships with delays, confirming that environmental and capacity constraints are primary delay drivers. **Load factor ($r = 0.14$)** shows a weak relationship, indicating limited impact from passenger volume, while **demand index ($r = 0.04$)** has negligible influence. The low inter-correlation among independent variables suggests minimal multicollinearity, making the findings statistically reliable. Additionally, a critical threshold was observed where **average delays exceeding 51 minutes significantly amplify perceived weather severity**, indicating nonlinear escalation in disruption effects.

Executive Recommendations

Based on the analysis, the following operational strategies are recommended:

- 1. Slot Optimization**
Prioritize early-morning departures, which consistently show **lower average delays ($\approx 33\text{--}35$ mins)** compared to night slots ($\approx 38\text{--}40$ mins).
- 2. Route-Based Risk Buffering**
Allocate additional buffer time and standby aircraft on high-risk routes such as **Hyderabad–Kolkata and Chennai–Kolkata**, where delays exceed **60 minutes under high weather severity**.
- 3. Seasonal Resource Planning**
Increase crew, ground handling, and maintenance buffers during **monsoon months (Jun–Sep)** and **winter (Nov–Dec)** when delays peak.
- 4. Congestion Management at Hub Airports**
Implement staggered scheduling and dynamic slot allocation at **Kolkata, Mumbai, and Hyderabad**, which show consistently higher congestion-driven delays.
- 5. Proactive Delay Control Threshold**
Trigger operational interventions once delays approach **50 minutes**, as delays beyond this point escalate rapidly due to compounded operational stress.